



CSC 309: Artificial Intelligence

Module 1: Foundations & Philosophy of AI

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1. Overview and Goals of Artificial Intelligence

Artificial Intelligence (AI) is a broad branch of computer science focused on building smart machines capable of performing tasks that typically require human intelligence. While humans use biological brains to process information, AI uses algorithms and computational models. The goals of AI have historically been divided into four quadrants (as defined by Stuart Russell and Peter Norvig), based on whether the focus is on **thought processes/reasoning** or **behavior/action**, and whether the standard of success is **human performance** or **ideal rationality**:

1. **Thinking Humanly (The Cognitive Modeling Approach):**
 - *Goal:* Building machines that simulate human internal thought processes.
 - *Application:* Cognitive science. We must determine how humans think (via introspection, psychological experiments, or brain imaging) and encode that into a program.
2. **Acting Humanly (The Turing Test Approach):**
 - *Goal:* Building machines that behave in a way that is observationally indistinguishable from humans.
 - *Focus:* Natural Language Processing, knowledge representation, and automated reasoning.
3. **Thinking Rationally (The "Laws of Thought" Approach):**
 - *Goal:* Building machines that use formal logic to derive the undeniably correct conclusions given available information.
 - *Limitation:* It is very difficult to take informal, real-world knowledge (which is often uncertain) and state it in formal logical terms.
4. **Acting Rationally (The Rational Agent Approach):**
 - *Goal:* Building machines that take the best possible action to achieve a specific goal in a given environment, maximizing expected utility.
 - *Note:* **This is the dominant approach in modern AI.** It is more general than the "laws of thought" approach because correct inference is just one of several possible mechanisms for achieving rationality.

AI Techniques

Traditional programming relies on algorithms (step-by-step instructions guaranteeing a solution). AI often employs **Heuristics**—rules of thumb that guide search and decision-making in complex spaces where exact algorithms are too computationally expensive (e.g., calculating every possible move in chess).

2. A Detailed History of AI

The journey of AI is characterized by cycles of massive optimism and funding, followed by periods of disappointment ("AI Winters").

- **The Genesis (1943–1956):**
 - 1943: Warren McCulloch and Walter Pitts propose the first mathematical model of a biological neuron.
 - 1950: Alan Turing publishes "Computing Machinery and Intelligence," introducing the Turing Test.
 - 1956: The term "Artificial Intelligence" is officially coined by John McCarthy at the Dartmouth Summer Research Project on Artificial Intelligence.
- **Early Enthusiasm & Great Expectations (1950s–1970s):**
 - Development of the General Problem Solver (GPS).
 - Creation of ELIZA (1966) by Joseph Weizenbaum, an early NLP program that simulated a psychotherapist.
 - *The Reality Check*: Researchers realized that scaling toy problems up to real-world problems led to a **combinatorial explosion** of computational states.
- **The First AI Winter (1974–1980):**
 - Funding dries up (e.g., the UK's Lighthill Report severely criticized AI research). Hardware limitations bottleneck progress.
- **The Expert Systems Boom (1980s):**
 - AI shifts focus from general-purpose problem solvers to narrow, domain-specific "Expert Systems" (e.g., R1/XCON used by DEC to configure computer systems).
- **The Second AI Winter (Late 1980s–Early 1990s):**
 - Expert systems proved too brittle, difficult to update, and expensive to maintain.
- **The Modern Era (1990s–Present):**
 - 1997: IBM's Deep Blue defeats world chess champion Garry Kasparov (a triumph of AI search algorithms).
 - 2012: The deep learning revolution begins when neural networks (AlexNet) crush traditional algorithms in the ImageNet computer vision competition.
 - Present: The era of Large Language Models (LLMs) and generative AI, driven by massive datasets and GPU computing power.

3. Types, Branches, and Applications of AI

Types of AI (Based on Capability)

- **Narrow/Weak AI (ANI - Artificial Narrow Intelligence):** AI trained to perform a single, specific task (e.g., facial recognition, spam filters, recommendation engines). **All AI existing today is Narrow AI.**

- **General/Strong AI (AGI - Artificial General Intelligence):** Hypothetical machines that possess human-level cognitive abilities across *all* domains. They can generalize knowledge from one domain to an entirely unrelated one.
- **Superintelligent AI (ASI - Artificial Superintelligence):** Hypothetical machines vastly smarter than the brightest human minds in practically every field, including scientific creativity and social intelligence.

Branches of AI

- **Machine Learning (ML):** Algorithms that parse data, learn from it, and make informed decisions without explicit programming.
- **Deep Learning (DL):** A subset of ML utilizing multi-layered artificial neural networks inspired by the human brain.
- **Natural Language Processing (NLP):** Giving machines the ability to read, understand, and derive meaning from human languages.
- **Expert Systems:** Rule-based AI that emulates the decision-making ability of a human expert (using an Inference Engine and a Knowledge Base).
- **Computer Vision:** Enabling machines to extract high-level, semantic understanding from digital images or videos.
- **Robotics:** The intersection of AI and mechanical engineering, allowing AI to interact with the physical environment.

Applications & Implications

- **Healthcare:** Disease prediction, automated radiology, drug discovery.
- **Finance:** Algorithmic trading, fraud detection, credit scoring.
- **Transportation:** Autonomous vehicles (self-driving cars), route optimization.
- **Advantages:** High precision, 24/7 availability, rapid processing of big data, automation of dangerous or monotonous tasks.
- **Disadvantages:** High cost of development, massive energy requirements, risk of encoding human biases into algorithms, lack of true emotional intelligence, and potential job displacement.

4. The Philosophy of Intelligence: Two Thought Experiments

When defining what makes a machine "intelligent," computer scientists and philosophers frequently debate two foundational thought experiments.

The Turing Test (Alan Turing, 1950)

Turing proposed a pragmatic, operational test to bypass the philosophical definition of "thinking."

- **The Setup:** A human evaluator sits in a room and has a text-based, typed conversation with two entities in another room—one is a human, the other is a machine.
- **The Goal:** If the evaluator cannot reliably tell which entity is the machine after a reasonable amount of time, the machine is said to have passed the test.
- **Requirements to Pass:** To succeed, the machine needs:
 1. *NLP* to communicate.
 2. *Knowledge Representation* to store what it knows.
 3. *Automated Reasoning* to use stored information to answer questions.
 4. *Machine Learning* to adapt to new circumstances in the conversation.
- (Note: The **Total Turing Test** adds video feeds and a physical hatch, requiring the machine to also possess *Computer Vision and Robotics*).

The Chinese Room Experiment (John Searle, 1980)

Philosopher John Searle introduced this argument to challenge the premise of Strong AI—specifically, the idea that passing the Turing Test equates to having a "mind" or true understanding.

- **The Setup:** Imagine John Searle, who speaks *only* English, locked in a room. He is given a massive rulebook written in English. This book contains instructions on how to manipulate Chinese symbols (e.g., "If you see symbol X, write down symbol Y").
- **The Execution:** Native Chinese speakers outside the room slide questions written in Chinese under the door. Searle uses his rulebook to find the matching symbols, writes down the required response symbols, and slides the paper back out.
- **The Argument:** To the people outside, the room appears perfectly fluent in Chinese; it easily passes the Turing Test. However, Searle *understands absolutely no Chinese*. He is simply manipulating symbols based on their syntax (shape/structure).
- **Significance:** Searle argues that computational models only achieve **syntax** (symbol manipulation), not **semantics** (actual meaning or understanding). Therefore, a computer running a program can only *simulate* intelligence, but it cannot truly *understand*.
- **The Systems Reply (Counter-argument):** Critics argue that while Searle himself doesn't understand Chinese, the *entire system* (Searle + the rulebook + the room) does possess understanding.

5. Optimal vs. Human-Like Behavior

To conclude the foundation of AI, we must distinguish between designing systems that act like

humans versus systems that act optimally.

- **Human-like Behavior:**
 - Humans are not strictly logical. We make decisions based on emotion, cognitive biases, fatigue, and heuristics.
 - Building an AI that acts human-like means programming these exact flaws and psychological traits.
 - *Applications:* Video game NPCs, psychological modeling, and human-computer interaction (HCI) where empathy simulation is required.
- **Optimal (Rational) Behavior:**
 - Doing the mathematically or logically "correct" thing to maximize the chances of achieving a specific goal, regardless of how a human would approach it.
 - Example: In chess, a human might play a suboptimal move to trick or psych out an opponent. A rational AI will always play the mathematically optimal move that maximizes its probability of a checkmate.
 - *Applications:* Autonomous vehicles, logistics planning, industrial automation.
- **Conclusion for CSC 309:** Modern AI engineering almost exclusively prioritizes building **Rational Agents** (Optimal Behavior) over mimicking human thought processes.